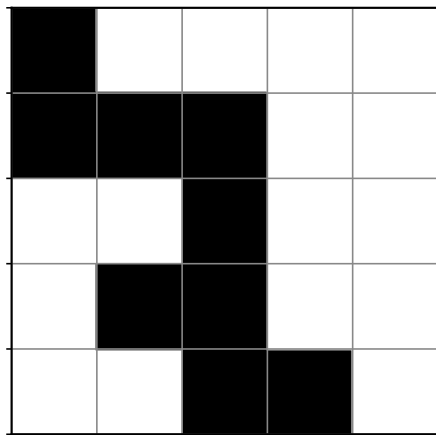
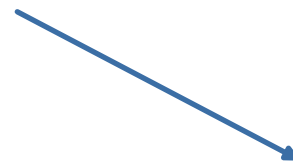
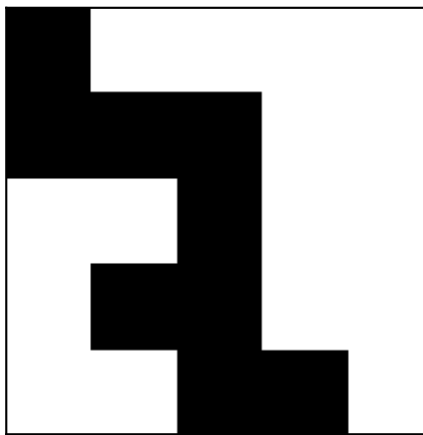


**ENCODE
(write)**

payload in a 5 x 5 grid
(each cell is 1 bit)



nearest-neighbor upsampling
bit map, 1 x H x W



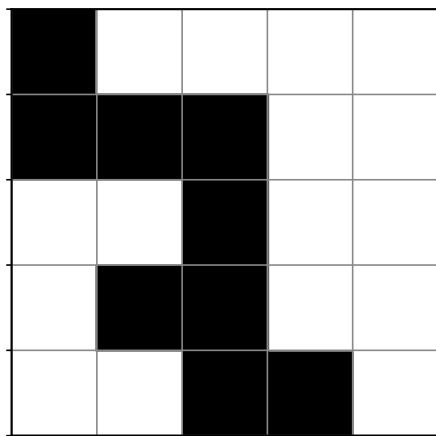
Stego QR
(encoder embeds the map into the cover)



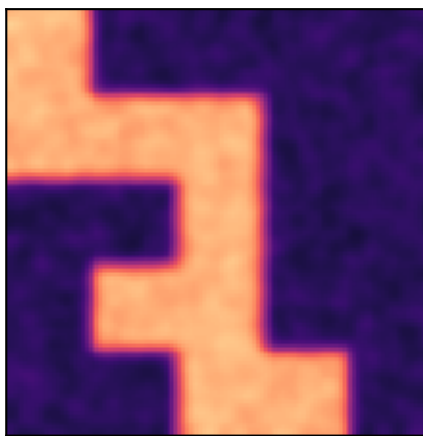
channel + decoder conv

**DECODE
(read)**

average pool to 5 x 5
read cells as bits



decoder activation map
1 x H x W



A shared convolution learns the same write and read operation at every cell. At 100 bits the spatial layout reached full recovery, while global broadcast remained near chance.